Part 2: Build with Claude Code

Video: Build ServiceNow Apps with AI — Claude Code Edition

> *We build a complete **AI Agent Management System** in ServiceNow using nothing but Claude Code in the terminal. No clicking through the ServiceNow UI. Just conversation → code → deploy.*

Prerequisites

From Part 1, you should have:

- ■ Node.js + NPM installed
- ■ Claude Code installed (`claude --version`)
- ■ ServiceNow SDK installed (`now-sdk --version`)
- ■ SDK authenticated to your instance (`now-sdk auth save`)

What We're Building

AI Agent Management System — Two tables connected to the CMDB:

Table	Key Fields
AI Agents	Name, Description, Status (Active/Inactive), Type (Autonomous/Assisted), Owner, Connected CI (→ CMDB)
Agent Tools	Name, Description, Type (Tool/Sub-Agent), Parent Agent (→ AI Agents), Capabilities

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Step 1: Create the Project

Open your terminal and create a new directory:

```
mkdir ai-agent-manager &amp;&amp; cd ai-agent-manager
```

Scaffold with the SDK

```
npx @servicenow/sdk init
```

When prompted:

- **App name:** AI Agent Manager
- **Scope:** `x_your_vendor_aiagent` (use your vendor prefix)
- **Description:** Manage AI agents, their tools, sub-agents, and CMDB connections

This creates the project structure with `now.config.json`, `package.json`, and `src/` folder.

Install Dependencies

```
npm install
```

Step 2: Launch Claude Code

```
claude
```

You're now in a conversational AI session that can see your entire project.

Step 3: Build the AI Agents Table

What to Tell Claude Code

Copy and paste this prompt (or say it in your own words):

```
Create a ServiceNow table for AI Agents using the Fluent API.

Table name: x_&lt;vendor&gt;_aiagent_ai_agent
Label: AI Agent

Fields:
- name (String, mandatory, max 200 chars) — the agent's name
- description (String, max 1000 chars) — what this agent does
- status (Choice: active, inactive, development, retired) — default "development"
- agent_type (Choice: autonomous, assisted, hybrid) — what kind of agent
- owner (Reference → sys_user) — who owns this agent
- connected_ci (Reference → cmdb_ci) — which CMDB asset it runs on
- version (String, max 50) — version number
- last_active (DateTime) — when the agent last ran
- documentation_url (URL) — link to docs

Put the file in src/fluent/tables/ai-agent.now.ts
```

What Claude Code Should Generate

Claude Code will create a file like this at `src/fluent/tables/ai-agent.now.ts`:

```
import {
  Table, StringColumn, ChoiceColumn, ReferenceColumn,
  DateTimeColumn, URLColumn
} from '@servicenow/sdk/core'

export const x_vendor_aiagent_ai_agent = Table({
  name: 'x_vendor_aiagent_ai_agent',
  label: 'AI Agent',
  schema: {
    x_vendor_aiagent_name: StringColumn({
      label: 'Name',
      maxLength: 200,
      mandatory: true
    }),
    x_vendor_aiagent_description: StringColumn({
      label: 'Description',
      maxLength: 1000
    }),
    x_vendor_aiagent_status: ChoiceColumn({
      label: 'Status',
      choices: {
        active: { label: 'Active' },
        inactive: { label: 'Inactive' },
        development: { label: 'In Development' },
        retired: { label: 'Retired' }
      },
      default: 'development'
    }),
    x_vendor_aiagent_agent_type: ChoiceColumn({
      label: 'Agent Type',
      choices: {
        autonomous: { label: 'Autonomous' },
        assisted: { label: 'Assisted' },
        hybrid: { label: 'Hybrid' }
      }
    }),
    x_vendor_aiagent_owner: ReferenceColumn({
      label: 'Owner',
      referenceTable: 'sys_user'
    }),
    x_vendor_aiagent_connected_ci: ReferenceColumn({
      label: 'Connected CI',
      referenceTable: 'cmdb_ci'
    }),
    x_vendor_aiagent_version: StringColumn({
      label: 'Version',
      maxLength: 50
    }),
```

```
      x_vendor_aiagent_last_active: DateTimeColumn({
        label: 'Last Active'
      }),
      x_vendor_aiagent_documentation_url: URLColumn({
        label: 'Documentation URL'
      })
    }
  })
```

Review & Fix

Look at what Claude Code generated. Check:

- ■ Table name matches your scope prefix
- ■ Column names start with your scope prefix
- ■ Choice values use object syntax `{ key: { label } }`, not arrays
- ■ `default` is used, not `defaultValue`
- ■ Export name matches table name exactly

If anything's wrong, just tell Claude Code: *"The scope prefix should be x_acme_aiagent, fix all the column names"*

Step 4: Build the Agent Tools Table

What to Tell Claude Code

```
Now create a second table for Agent Tools / Sub-Agents.

Table name: x_&lt;vendor&gt;_aiagent_agent_tool
Label: Agent Tool

Fields:
- name (String, mandatory, max 200) — tool or sub-agent name
- description (String, max 1000) — what it does
- tool_type (Choice: tool, sub_agent, api, plugin) — what kind of component
- parent_agent (Reference → x_&lt;vendor&gt;_aiagent_ai_agent, mandatory) — which agent owns thi
- capabilities (String, max 2000) — detailed capabilities description
- permissions (String, max 500) — what permissions/access it has
- is_active (Boolean) — whether this tool is currently enabled
- execution_count (Integer) — how many times it has been invoked

Put it in src/fluent/tables/agent-tool.now.ts
```

What Claude Code Should Generate

A similar Fluent API file with `BooleanColumn`, `IntegerColumn`, and a `ReferenceColumn` pointing to the AI Agents table.

Step 5: Add ACLs (Security)

What to Tell Claude Code

```
Create ACLs for both tables. Requirements:
- Admin role: full CRUD on both tables
- itil role: read + write on both tables
- Any authenticated user: read only

Put them in src/fluent/acls/agent-acls.now.ts
```

Step 6: Build and Deploy

Build the App

Tell Claude Code:

```
Run the build command: now-sdk build
```

Or run it yourself:

```
npm run build
```

Check for Errors

If the build fails, paste the error back to Claude Code:

```
The build failed with this error: [paste error here]. Fix it.
```

Common issues Claude Code might hit:

- Wrong scope prefix → tell it the correct one
- `ChoiceColumn` using array instead of object → it'll fix the syntax
- Missing imports → it'll add them

Deploy to Your Instance

```
now-sdk install --auth mydev
```

This builds, packages, and deploys the app to your ServiceNow instance.

Step 7: Verify in ServiceNow

1. Log into your ServiceNow instance

2. Navigate to **AI Agent Manager** in the application navigator

3. You should see:

• **AI Agents** — list view with all your fields

• **Agent Tools** — list view linked to agents

4. Create a test record:

• Add an AI agent named "Document Parser"

• Set status to "Active", type to "Autonomous"

• Link it to a CI in your CMDB

• Add a tool called "PDF Extractor" linked to this agent

Claude Code Tips & Tricks

Useful Prompts During Development

What You Want	What to Say
See project structure	*"Show me the project file structure"*
Fix a build error	*"The build failed with [error]. Fix it"*
Add a new field	*"Add an 'email' field to the AI Agents table"*
Create a business rule	*"Create a business rule that sets last_active to now() when status changes to active"*
Explain something	*"What does this ReferenceColumn do?"*
Run a command	*"Run now-sdk build and show me the output"*

What Claude Code Does Well

• Generates complete, valid Fluent API code

• Understands ServiceNow scope prefixes

• Can read and fix build errors

• Runs terminal commands for you

• Keeps context of the entire project

What to Watch For

- Always verify the scope prefix matches your `now.config.json`
- Double-check Reference fields point to the right table
- Review generated code before deploying — don't blindly deploy

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Summary

Step	What We Did
1	Created project with `npx @servicenow/sdk init`
2	Launched Claude Code in the project
3	Built AI Agents table via conversation
4	Built Agent Tools table via conversation
5	Added ACLs via conversation
6	Built and deployed with `now-sdk install`
7	Verified in ServiceNow

Total time writing code manually: 0 lines. Claude Code wrote everything.

Next: [Part 3 — Build with GitHub Copilot →]